Predictive Modeling and Explainable Machine Learning Analysis of Social Media Addiction

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Supervised by

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DECLARATION

I hereby declare that the dissertation that I have submitted "Predictive Modeling and Explainable Machine Learning Analysis of Social Media Addiction" to Dublin Business School for the award of master's in Data Analytics under the guidance of supervision of Mr. Jagadeeswaran Thangaraj is solely the result of my own work; collaboration contributions have been acknowledged and are explicitly referenced in the text. This work has not been submitted to any university or college for the award of Degree.

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DEDICATION

This thesis is dedicated, first and foremost, to my father, Engr Celestine Shausu whose steadfast guidance and quiet sacrifices have shaped the person I have become. His unwavering confidence in my abilities has been a constant source of strength throughout this journey, and it is to him that I owe the deepest measure of gratitude.

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To my siblings, Hembafan, Faeren and Doose Shausu for their steadfast support and the sense of belonging they have given me, I am equally grateful. Their presence has been a source of comfort and perspective throughout this endeavour.

It is with profound gratitude and love that I dedicate this work to them.

This work is as much mine as it is yours.

ABSTRACT

The increasing prevalence of social media use has increased concerns regarding its problematic and addictive engagement patterns, yet most existing research relies on correlational and regression-based methods that struggle to capture the complex, non-linear relationships between behavioural and psychological predictors. This study developed and evaluated machine learning models for predicting social media addiction risk, and applied explainable AI to identify the variables driving those predictions. Using a publicly available dataset of 258 adolescents and young adults, a binary "at-risk" target was constructed from the Bergen Social Media Addiction Scale (BSMAS) using a validated clinical cut-off, producing a severely imbalanced dataset (88.8% not-at-risk, 11.2% at-risk). Four classifiers, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine, were each combined with three imbalance-handling strategies (no rebalancing, class-weighting, and SMOTE), and evaluated using nested, repeated, stratified cross-validation against PR-AUC and F1. Random Forest achieved the highest PR-AUC (0.232), narrowly ahead of SVM with SMOTE (0.231), while Decision Tree achieved the highest F1 (0.215). On a held-out test set, the selected Random Forest model achieved a PR-AUC of 0.361 and ROC-AUC of 0.676, both well above the 0.115 baseline expected by chance, though recall at the default threshold remained low (0.167). SHAP analysis of the final model showed that psychometric variables: cognitive interference from smartphone use, trait and state anxiety, and self-esteem contributed more to predictions than behavioural usage variables such as time spent or number of platforms used. These findings suggest that machine learning can meaningfully outperform chance-level prediction of social media addiction risk, but that reliable case-level identification remains constrained by small sample size and severe class imbalance. The results further suggest that screening and intervention efforts may benefit from incorporating psychometric indicators of psychological vulnerability alongside, or in place of, behavioural usage metrics. Recommendations for future research include larger and multi-site samples, longitudinal designs, and external validation across cultural contexts.

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1. INTRODUCTION

1.1 Background of Study

Over the past decade, social media platforms have transformed communication patterns globally, with billions of users engaging daily across multiple digital platforms. As of early 2024, about 66.67% of the global population are active social media users (Jiang, 2023; Liţan, 2025). This expansion is characterized by a rapid rate of adoption, with an estimated 266 million new users joining digital platforms in 2023 alone (Jiang, 2023). Digital penetration has reached a critical point, that as of 2020 it was indicated that there are over 4.3 billion social media users worldwide (Gambo and Özad, 2020). The intensity of engagement is particularly high among younger people; in developed countries where internet access is nearly universal, many young adults are active users (Thiagarajan, Newson and Swaminathan, 2025). A study in the US in 2025, showed that the daily time spent on these platforms averages over 4 hours for about 50.4% of teenagers. This growth is mainly caused by widespread mobile internet access, which has made more people constantly connected; and moved news and communication mostly online (Butler, 2024; Castelo *et al.*, 2025).

Because of the rapid increase in social media usage, concerns have emerged regarding problematic and addictive patterns of engagement. These patterns are classified under the umbrella of behavioral addictions, which, unlike substance-use disorders, act on the human organism indirectly by stimulating the dopaminergic reward system through compulsive digital interaction (Semolini *et al.*, 2025). Within this framework, social media addiction is viewed as a specific "Internet-related disorder" where the user develops a psychological dependence on applications to regulate mood or achieve social validation (Geisel, Lipinski and Kaess, 2021). Scholars categorize these behaviors as "non-substance addictions," encompassing both mobile phone and internet-related dependencies that mirror the diagnostic criteria of traditional addiction, such as withdrawal, conflict, and relapse (Zou *et al.*, 2017). Even though the academic discourse remains divided on the validity of these classifications, the clinical impact is significant; frequent, compulsive checking of notifications is often linked to co-occurring psychological disorders, including anxiety, depression, and ADHD (Semolini *et al.*, 2025). As digital platforms are designed to maximize user retention through intermittent reinforcement schedules, the transition from functional use to "information overload" or "cyber-relational addiction" represents a growing public health priority (Kuss and Griffiths, 2017; Zou *et al.*, 2017).

Studies link excessive social media use with negative psychological, academic, and social outcomes (Vannucci, Flannery and Ohannessian, 2017). Others involving early adolescents indicate a unidirectional link where increases in social media consumption directly precede elevated depressive symptoms in subsequent years (Nagata *et al.*, 2025). This relationship is often mediated by the Fear of Missing Out (FoMO) and a Preference for Online Social Interactions (POSI), which transforms casual browsing into a maladaptive coping mechanism for those who already experience anxiety (Bunker *et al.*, 2024). The impact is particularly pronounced in academic environments; for instance, a staggering 74.34% of medical students in one study were found to be at risk of social media addiction, with such problematic use serving as a significant predictor of depressive severity (Sserunkuuma *et al.*, 2023). Network analysis further reveals that "sadness" and "cognitive preoccupation" act as central bridge symptoms, locking users into a cycle where smartphone addiction and depression reinforce one another (Gao *et al.*, 2025). Beyond mood disorders, the displacement of sleep and productive study time contributes to a decline in overall functional well-being, as students often turn to these platforms to escape existing stressors, only to find their academic performance and sleep quality further compromised (Mengistu *et al.*, 2023).

To systematically assess addictive patterns of social media use, researchers have developed validated psychometric instruments such as the Bergen Social Media Addiction Scale (BSMAS), Social Media Disorder Scale (SMD), and related behavioral assessment tools. These tools typically employ a Likert-type scoring system to quantify the severity of behaviors across multiple dimensions, most notably functional deterioration, control difficulty, and social isolation (Ayğar and Uzun, 2018). These Social Media Addiction Scales have demonstrated robust psychometric properties, indicating high internal consistency across both adolescent and adult populations (Gökdaş and Kuzucu, 2019). Construct validity for these 10-item measures is frequently established through Confirmatory Factor Analysis (CFA), which verifies that the items accurately represent the underlying theoretical structure of addiction, such as withdrawal and preoccupation (Escalera-Chávez, Rojas-Kramer and Larracilla-Salazar, 2026). While some shorter scales, like the 9-item Social Media Disorder (SMD) scale, focus on a one-factor structure for rapid screening (Van Den Eijnden, Lemmens and Valkenburg, 2016), the 10-item formats are often preferred for their ability to capture a more nuanced three-factor model while remaining brief enough for clinical and field research (Gökdaş and Kuzucu, 2019). Despite their widespread use, some studies suggest that internal consistency can vary based on the specific demographic, with alpha coefficients occasionally dipping in younger adolescent

samples, indicating a need for ongoing age-specific validation (Watson, Prosek and Giordano, 2020).

Most empirical investigations rely primarily on descriptive statistics and traditional regression approaches. These methods are fundamentally limited because they emphasize association rather than prediction, failing to account for the complex, non-linear interactions inherent in behavioral data. A significant portion of current research is criticized for its heavy reliance on self-report questionnaires, which are susceptible to social desirability bias and are often insufficient for formal clinical diagnosis (Panova and Carbonell, 2022). Furthermore, the field suffers from a lack of longitudinal and experimental designs, leaving the long-term stability of social media addiction and its potential for false positives in screening largely unverified (Carbonell and Panova, 2017).

The advancement of machine learning techniques presents an opportunity to move beyond correlation-based analysis toward predictive and interpretable risk modeling. Recent research shows that machine learning is increasingly used to predict mental health conditions because of the growing availability of behavioral and psychological data (Chung and Teo, 2022). Several studies demonstrate that machine learning models can be applied to predict disorders such as depression, anxiety, and post-traumatic stress disorder (Chung and Teo, 2022). Similarly, machine learning algorithms such as Support Vector Machines and k-Nearest Neighbor have been successfully used for the detection and prognosis of mental health disorders (Andrew *et al.*, 2023). Empirical research also shows that algorithms such as Random Forest and adaptive boosting can effectively predict mental well-being among university students using behavioral data (Jain *et al.*, 2021; Rahman *et al.*, 2023). However, many machine learning models operate as "black boxes," making their predictions difficult to interpret. To address this challenge, researchers have applied Explainable Artificial Intelligence techniques such as SHAP to improve transparency and understanding of mental health prediction models (Nordin *et al.*, 2023; Atlam *et al.*, 2025).

1.2 Problem Statement

Concerns about problematic and addictive patterns of social media use have grown alongside its increasing prevalence. While a substantial body of research has examined the relationship between social media use and psychological outcomes, most of this work relies on traditional statistical techniques that emphasize correlation over prediction. Few studies, moreover,

incorporate explainable machine learning approaches capable of offering interpretable insight into the behavioural and psychological factors that drive addictive use. This gap, the lack of models that are both predictive and interpretable, limits the development of targeted interventions and evidence-based digital wellbeing strategies. There is therefore a need to apply more advanced analytical techniques to existing data in order to develop predictive models of social media addiction risk that are both accurate and explainable.

1.3 Research Questions

This study seeks to answer the following research questions:

1. Can machine learning models accurately predict social media addiction risk?
2. Which variables contribute most significantly to addiction risk predictions?

1.4 Aim

The aim of this study is to develop and evaluate predictive and explainable machine learning models for identifying social media addiction risk using a publicly available social media addiction dataset.

1.5 Objectives

The objectives of this study are as follows:

1. To identify key behavioral and demographic factors associated with social media addiction.
2. To develop predictive models using traditional statistical and machine learning techniques.
3. To compare the performance of the developed models using appropriate classification metrics.
4. To apply explainable AI techniques to interpret model predictions.

1.6 Significance of Study

The increasing prevalence of social media use has increased concerns regarding its problematic and addictive engagement patterns. Although there are numerous studies examining the associations between social media usage and psychological outcomes, most of them rely on traditional statistical techniques that emphasize correlation rather than prediction. Furthermore, existing studies rarely incorporate explainable machine learning approaches that provide interpretable insights into behavioral determinants. The absence of predictive and interpretable models limits the development of targeted interventions and data-driven digital wellbeing

strategies. Therefore, there is a need to apply advanced analytical techniques to publicly available behavioral social media addiction datasets to develop accurate and explainable predictive models of addiction risk.

1.7 Definition of Terms

Social Media Addiction

A pattern of excessive and compulsive social media use characterized by impaired control, preoccupation, and negative consequences.

Bergen Social Media Addiction Scale

The Bergen Social Media Addiction Scale (BSMAS) is a 6-item self-report tool used to measure the severity of problematic social media use.

Predictive Modeling

The use of statistical or machine learning algorithms to forecast outcomes based on input variables.

Machine Learning

A subset of artificial intelligence that enables systems to learn patterns from data without explicit programming.

Explainable Artificial Intelligence (XAI)

Techniques that make machine learning model decisions interpretable to humans.

Risk Profiling

The classification of individuals based on their likelihood of exhibiting a particular condition or behavior.

2. LITERATURE REVIEW

2.1 Behavioural Addiction Theory

Behavioral addiction refers to compulsive engagement in rewarding non-substance related activities despite negative consequences (Alavi *et al.*, 2012). They involve functional impairments and/or significant distress generated by excessive engagement in non-substance-related behaviors that are perceived as rewarding and/or relieving, but which are difficult for the affected individual to control. These behaviors receive increasing priority in everyday life and are continued despite negative consequences (Alavi *et al.*, 2012; Brand *et al.*, 2025). The key difference between substance addiction and behavioural addiction is that substance addiction involves compulsive use of chemicals (drugs/alcohol) that alter brain chemistry, leading to physical dependence, tolerance, and severe withdrawal symptoms while behavioral addiction involves compulsive actions (gambling, gaming, shopping) that trigger the brain's reward system without physical substance ingestion, often resulting in psychological, rather than physical, withdrawal (Alavi *et al.*, 2012). Building on this distinction, (Griffiths, 2005) argues that these diverse behaviors share a core pathological structure. This perspective suggests that all addictions, regardless of whether they involve a drug, consist of six distinct common components: salience, mood modification, tolerance, withdrawal, conflict, and relapse. By viewing these excessive behaviors as part of a unified biopsychosocial process, researchers suggest that a flexible approach is the most practical way forward (Griffiths, 2005). This provides a framework for understanding how excessive engagement in digital behaviors may evolve into addictive patterns.

Several psychological mechanisms have been proposed to explain how behavioral addictions develop and persist over time. Central to these processes is the brain's reward system, where excessive behaviors enhance dopaminergic synaptic function in the ventral striatum to create a reinforcing "high" (Grassi, Cecchelli and Vignozzi, 2022). This dopamine release in the nucleus accumbens reinforces the behavior, driving a cycle of reinforcement learning where the individual seeks to repeat the rewarding action (İzci, Usta Sağlam and Ergelen, 2022). Over time, this shifts from goal-directed behavior to habit formation, often characterized by impaired decision-making and heightened impulsivity (Grassi, Cecchelli and Vignozzi, 2022). Midbrain dopamine signaling further complicates this by facilitating reward prediction, which creates a compulsive loop of anticipation and engagement (Brand *et al.*, 2025). Digital platforms exploit these pathways by offering instant gratification, which short-circuits traditional self-regulation

and locks the user into a reward-seeking state (Termann, 2025). These mechanisms are particularly relevant in the context of digital technologies that are designed to maximize user engagement.

2.2 Social Media Addiction Studies

The rapid growth of social media platforms has significantly transformed communication, information sharing, and social interaction across the world. Recent data indicates a massive surge in global adoption, with active user numbers rising from 2.73 billion in 2017 to over 5 billion in 2024, and projections suggesting a reach of 6 billion by 2027 (Labudová, 2024; Marzo *et al.*, 2024). This expansion is led by "social networking giants" such as Facebook, which maintains over 3 billion active users, followed by YouTube and the dual-platform dominance of WhatsApp and Instagram (Gambo and Özad, 2020; Labudová, 2024). This growth is not merely numerical but structural; platforms like WhatsApp have become the uncontested leaders in global messaging across 180 countries, while others like Pinterest and Twitter have carved out significant market shares of 10.45% and 14.33% respectively (Gambo and Özad, 2020; Бочко, Кузяк and Aleksandraviciute, 2020).

Engagement metrics further illustrate this integration, with the average worldwide daily social media use estimated between 2.24 and 2.5 hours; a 30-minute increase since 2015 (Pellegrino, Stasi and Bhatiasevi, 2022; Marzo *et al.*, 2024; Montag *et al.*, 2024). However, objective smartphone tracking suggests these figures may be higher in specific contexts, with some studies recording users spending over 3 hours and 22 minutes daily on social applications (Coyne, Voth and Woodruff, 2023). This intensity is particularly pronounced among youth demographics; younger users exhibit significantly longer smartphone usage durations than older cohorts, with adolescents frequently engaging in mobile social media for substantial portions of their day (Almourad *et al.*, no date; Hall and Liu, 2022). However, the increasing integration of social media into everyday life has also raised concerns regarding excessive and problematic usage.

To assess addictive patterns of social media use, several psychometric instruments have been developed and validated by researchers. The systematic assessment of addictive social media patterns through multiple psychometric instruments have been developed to assess problematic social media use, each differing in dimensional structure and scoring methodology. While traditional 10-item instruments are often favored for capturing a nuanced three-factor model,

specifically addressing functional deterioration, control difficulty, and social isolation (Ayğar and Uzun, 2018; Gökdaş and Kuzucu, 2019); researchers must account for significant structural variations across the field. For instance, while the Bergen Social Media Addiction Scale (BSMAS) is frequently cited alongside 10-item tools, it often functions as a 6-item unidimensional measure; Tung et al. (2022) validated this shorter structure using Rasch analysis and confirmed its strong correlation with psychological distress in university samples. However, qualitative "think-aloud" protocols by Schmidt et al. (2022) caution that even widely used tools like the BSMAS can suffer from outdated or abstract wording that may impede content validity among younger users. This push for greater precision has led to more expansive models, such as the 22-item Indian Social Media Addiction Scale, which utilizes a six-factor framework; Salience, Tolerance, Mood Modification, Relapse, Withdrawal, and Conflict to map the behavioral complexity of addiction in specific cultural contexts (Saqib and Amin, 2024). Furthermore, contemporary 10-item adaptations like the SOMEDIS-A/-P have moved toward clinical standardization by aligning directly with ICD-11 criteria for Gaming Disorder (Paschke *et al.*, 2023). Despite the robust construct validity often established via Confirmatory Factor Analysis (Escalera-Chávez, Rojas-Kramer and Larracilla-Salazar, 2026), the tendency for internal consistency to fluctuate in younger adolescent cohorts (Watson, Prosek and Giordano, 2020) suggesting that analysis must remain sensitive to age-specific reliability and potential item-level limitations and move beyond traditional statistical analysis.

2.3 Predictive Modelling in Psychology

Psychological research has traditionally relied on statistical techniques such as correlation analysis, regression models, and hypothesis testing to examine behavioral relationships. In the past, tools like ANOVA and linear regression have served as the backbone of data analysis, particularly for identifying associations among variables and conducting mediation or moderation analyses (Hayes and Rockwood, 2017; Blanca, Alarcón and Bono, 2018). While these parametric tests are effective for small sets of theoretically conceived variables, they face significant constraints when applied to the large-scale digital datasets now common in the field (Lavelle-Hill, Smith and Murayama, 2025). Furthermore, traditional structural equation models and regression-based approaches often struggle with issues like multicollinearity and violations of data independence, which can lead to unstable parameter estimation (Lavelle-Hill, Smith and Murayama, 2025).

In recent years, predictive analytics has gained increasing attention in psychological research as a tool for forecasting behavioral outcomes. This shift addresses a critical limitation in the traditional approach, where models often demonstrate high "goodness of fit" to existing data but fail to accurately predict future behaviors or generalize to out-of-sample data (Yarkoni and Westfall, 2017). Unlike the data-modeling approach of classical statistics, predictive modeling leverages algorithmic frameworks that make fewer assumptions about data-generating mechanisms, making them ideal for risk prediction and early detection systems in mental health (Lavelle-Hill, Smith and Murayama, 2025). By prioritizing predictive accuracy over mechanistic explanation, researchers can better account for complex, non-linear patterns that traditional linear models might overlook. This shift toward predictive analysis has opened new opportunities for applying machine learning techniques to behavioral data.

2.4 Machine Learning in Behavioral Analytics

Machine learning has emerged as a powerful analytical approach for identifying patterns and predicting outcomes in large behavioral and psychological datasets. In the healthcare sector, artificial intelligence is increasingly utilized to refine diagnoses and monitor patient conditions, providing clinicians with data-driven insights that traditional tools may overlook (Andrew *et al.*, 2023). By analyzing diverse data modalities, including electronic health records, wearable devices, and social media monitoring; researchers can develop early detection systems for mental illness that significantly improve patient outcomes (Islam *et al.*, 2024). For instance, models integrated into clinical workflows have demonstrated the ability to continuously monitor and predict the risk of mental health crises, assisting professionals in managing caseloads and mitigating risks in over 60% of cases (Garriga *et al.*, 2022). Consequently, researchers have increasingly adopted machine learning models to analyze behavioral patterns and predict mental health outcomes.

Several machine learning algorithms have been widely applied in behavioral analytics due to their ability to capture complex nonlinear relationships within data. Random Forest (RF) and Gradient Boosting (GB) are frequently employed for their high accuracy in classifying psychological distress and identifying overlapping mental symptoms (Warda *et al.*, 2017; Chen *et al.*, 2022). Additionally, Support Vector Machines (SVM) have proven effective in predicting psychosis disorders by classifying multimodal data into high-risk or normal categories, while Artificial Neural Networks (ANN) often provide superior predictive effects for binary classification tasks (Chen *et al.*, 2022; Andrew *et al.*, 2023). Recent developments

even utilize ensemble voting classifiers to improve the identification of disorders in specific populations, such as autistics and young children (Atlam *et al.*, 2025). Despite their strong predictive performance, many of these models are often criticized for their limited interpretability.

To address the interpretability challenge associated with machine learning models, researchers have increasingly adopted Explainable Artificial Intelligence techniques. These methods aim to solve the "black box" problem, where high-performing deep learning and ensemble models, such as BiLSTM or XGBoost, produce accurate predictions without revealing the underlying logic of their decision-making process (Kerz *et al.*, 2023; Mokheleli, Bokaba and Mbunge, 2025). By integrating tools like SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), researchers can now achieve both global and instance-level transparency. For example, SHAP can rank feature importance to show that variables like "past mental health disorders" or "treatment-seeking behavior" are primary drivers of workplace mental health outcomes, while LIME provides case-specific justifications for individual predictions (Byeon, 2023; Mokheleli, Bokaba and Mbunge, 2025). This is especially critical in clinical settings where relying solely on standard metrics like accuracy can be misleading if the model focuses on irrelevant or unsound features (Pendyala and Kim, 2024). These techniques enhance transparency, allowing researchers to understand how different variables influence predictive outcomes.

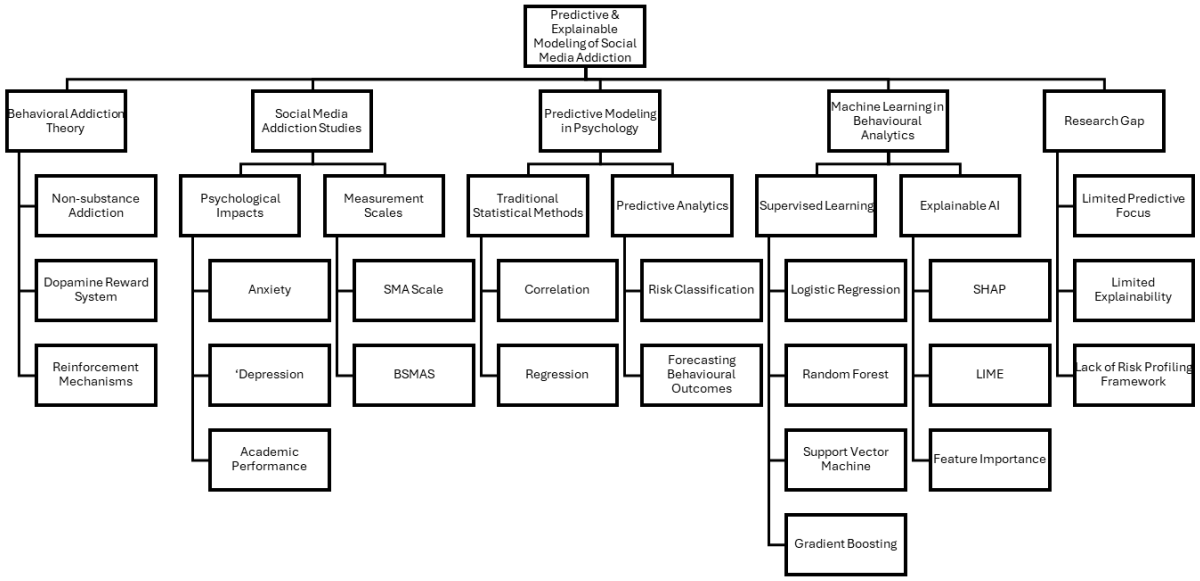


Figure 1 The increasing prevalence of social media use has increased concerns regarding its problematic and addictive engagement patterns

2.5 Research Gap

Despite the expanding literature on social media addiction, several methodological limitations remain. Most studies rely on cross-sectional surveys and self-report instruments (Jones *et al.*, 2024; Alluhidan *et al.*, 2026). These descriptive analyses capture broad usage trends but cannot record moment-to-moment behavior or establish causality. Few works have used predictive modeling techniques; instead, research often reports correlations or group differences without testing how well models can forecast who will become addicted. As a result, current research provides limited insight into the ability to accurately predict individuals who are at high risk of social media addiction.

Furthermore, relatively few studies have integrated predictive machine learning models with explainable AI techniques in the context of social media addiction research. Existing ML-based addiction studies tend to focus on maximizing prediction accuracy while treating models as "black boxes," paying limited attention to interpretability (Liu *et al.*, 2025). This means that even when high-risk cases are identified, we do not know which features drove those predictions. By contrast, explainable methods like SHAP and LIME can highlight which

factors (e.g. anxiety, stress) contribute most to an individual's risk score. For example, SHAP analyses in related work on Internet addiction have revealed consistent, interpretable feature contributions to each prediction (Liu *et al.*, 2025; Kong *et al.*, 2026). Such interpretable models would enable personalized risk profiling and data-driven interventions, by making clear why a person is classified as high-risk. Therefore, this study seeks to address this gap by applying predictive and explainable machine learning techniques to the dataset in order to identify key factors associated with social media addiction risk.

3. RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the methodological framework adopted to develop predictive and explainable models for identifying social media addiction risk using the Social Media Addiction dataset. The study integrates statistical analysis, supervised machine learning, and explainable artificial intelligence (XAI) techniques to ensure both predictive accuracy and interpretability. The methodology is structured into six major stages: data preprocessing, exploratory data analysis, feature engineering, model development, model evaluation, and model explainability. The overall objective is to construct a robust, data-driven risk classification framework capable of identifying individuals at varying levels of social media addiction.

3.2 Research Design

This study adopts quantitative research design grounded in supervised machine learning framed as a binary classification problem. The dependent variable represents levels of addiction severity derived from the BSMAS scale using its validated clinical cut-off (BSMAS ≥ 19 = "at-risk"; BSMAS < 19 = "not at-risk"). Unlike traditional explanatory studies that focus on statistical associations (Jones *et al.*, 2024; Alluhidan *et al.*, 2026), this research emphasizes predictive modeling. The goal is not only to determine which variables are related to addiction but also to develop models capable of accurately classifying individuals into risk categories. The research workflow follows a structured analytical pipeline: (i) Data Preprocessing (ii) EDA (iii) Feature Engineering (iv) Baseline Establishment (v) Model Training with Imbalance Handling Comparison (vi) Nested Cross-Validation and Model Selection (vii) Final Held-Out Evaluation (viii) Explainability Analysis. Figure 2 shows an illustration of the research design flow.

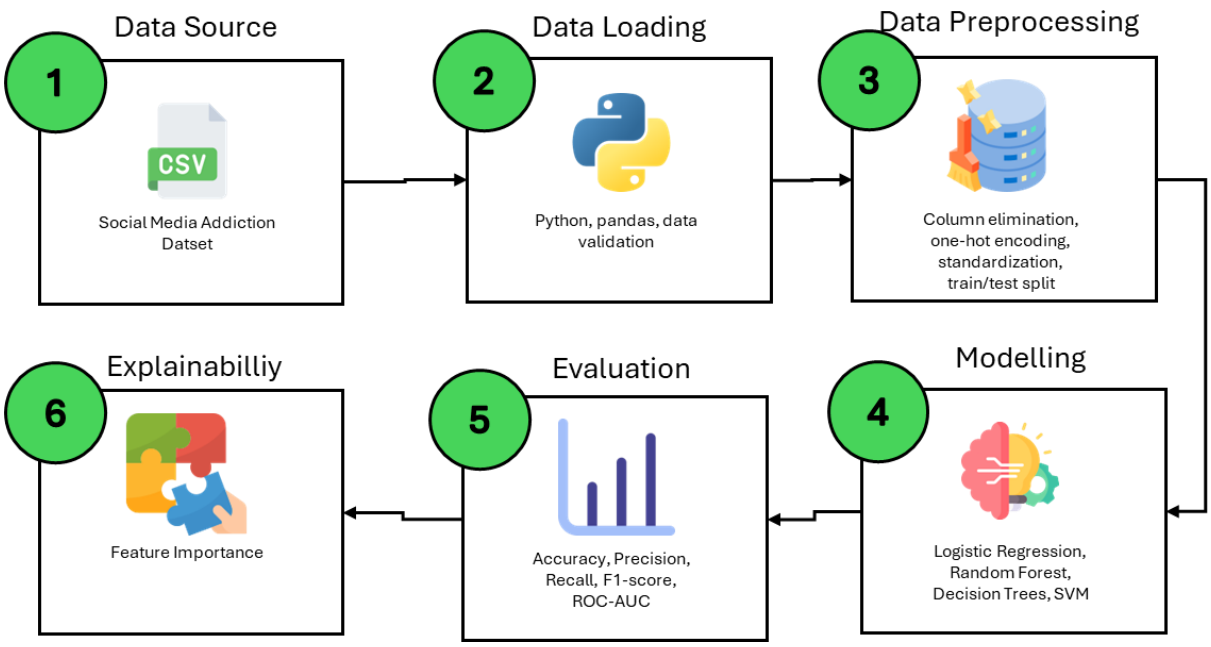


Figure 2 The research design workflow

3.3 Dataset

The dataset used in this study was obtained from the Mendeley Data repository and consists of survey responses examining social media use and psychological well-being among adolescents and young adults. The dataset contains 258 observations and 43 variables (with some duplicate variables), capturing demographic characteristics, patterns of social media engagement, gaming behavior, and standardized psychometric scale scores. Participants' demographic information includes sex, age, and school failure history. Social media usage variables cover platform preference (e.g., WhatsApp, Facebook, Instagram, Twitter, Snapchat, TikTok, and others), daily usage frequency, time spent on social media, primary device used, and behavioral tendencies such as compulsive checking and perceived time distortion while online. Several columns appear as coded or auxiliary variables corresponding to categorical responses.

In addition to behavioral indicators, the dataset incorporates validated psychological assessment scales to measure addiction risk and emotional well-being. These include the Bergen Social Media Addiction Scale (BSMAS), which assesses addictive social media tendencies, alongside a cut-off indicator for identifying high-risk users. The dataset also contains scores from the Rosenberg Self-Esteem Scale (RSES), the State-Trait Anxiety Inventory (STAI Y-1 for state anxiety and STAI Y-2 for trait anxiety), and the CSIQ-A scale. These instruments provide quantitative measures of self-esteem, anxiety levels, and related

psychosocial constructs. The integration of behavioral metrics with psychometric scales makes the dataset suitable for predictive modeling, correlation analysis, and explainable AI applications aimed at understanding the determinants and consequences of social media addiction. Table 1 provides a summary of variables in the dataset including their description and data type.

Table 1 Summary of variables

Variable Name	Description	Data Type	Role in Study
Sex	Participant gender	Categorical	Demographic predictor
Age	Participant age (years)	Numerical (Continuous)	Demographic predictor
School failure	History of school failure	Categorical (Binary)	Demographic predictor
Whatsapp / Facebook / Instagram / Twitter / Snapchat / TikTok / Other social	Most frequently used platform (WhatsApp, Instagram, TikTok, etc.)	Categorical	Behavioral predictor
Daily use of social media	How often social media is accessed daily	Ordinal	Behavioral predictor
Time spent on social medua	Time spent daily on social media	Numerical / Ordinal	Behavioral predictor
Which device	Primary device for access (phone, tablet, PC)	Categorical	Behavioral predictor
SM Checking	Urge to frequently check social media	Ordinal	Behavioral indicator
Time's flow	Losing track of time while online	Ordinal	Behavioral indicator
Video-games and online play	Whether participant plays digital games	Categorical (Binary)	Behavioral predictor
Time spent online playing videogame	Frequency of gaming	Ordinal	Behavioral predictor
BSMAS	Bergen Social Media Addiction Scale total score	Numerical (Continuous)	Target / Outcome variable
BSMAS cutoff	Addiction risk classification (based on cut-off score)	Binary	Target (Classification)
RSES	Rosenberg Self-Esteem Scale score	Numerical (Continuous)	Psychological variable
STAI Y-1	State Anxiety score	Numerical (Continuous)	Psychological variable
STAI Y-2	Trait Anxiety score	Numerical (Continuous)	Psychological variable
CSIQ-A	CSIQ-A scale score	Numerical (Continuous)	Psychological variable

3.4 Data Preprocessing

Data preprocessing is a critical step to ensure the reliability and quality of predictive models. Raw datasets often contain inconsistencies, scaling differences, or structural issues that can negatively impact model performance.

3.4.1 Handling Missing Values

The dataset will be examined for missing entries using descriptive statistics and null value detection techniques. Where missing values are negligible, listwise deletion will be applied. No missing data was identified, so no deletion or imputation procedures were required for this dataset.

3.4.2 Feature Encoding

Only the two nominal categorical variables, Sex and Which device, were one-hot encoded (with the first category dropped to avoid collinearity), since treating them as raw integers would impose a false ordinal structure on distance-sensitive models such as Logistic Regression and SVM. All remaining ordinal behavioral variables (e.g., daily usage frequency, compulsive checking, time distortion) were retained in their existing integer-coded form and standardized directly, since converting them to dummy variables would discard their inherent ordering .

3.4.3 Normalization and Scaling

Numeric features were standardized using z-score normalization (StandardScaler) exclusively. Scaling was performed inside a ColumnTransformer within the modelling pipeline structure so that scaling parameters were fit only on training data within each cross-validation fold, preventing information leakage from validation or test folds.

$$z = \frac{(x - \mu)}{\sigma}$$

3.4.4 Feature Engineering

Two features were engineered; n_platforms , was constructed as the sum of the seven binary platform-use indicators (Whatsapp, Facebook, Instagram, Twitter, Snapchat, TikTok, Other social), capturing the breadth of platform use as a single numeric predictor. Also the target variable, addiction_risk is derived from the Binary at-risk flag, BSMAS ≥ 19.

3.5 Exploratory Data Analysis (EDA)

3.5.1 Distribution of Addiction Scores

The distribution of addiction risk scores were examined using histograms and boxplots. Measures of central tendency (mean, median) and dispersion (standard deviation, range) were computed. This analysis helped determine the overall severity trend and potential skewness within the population.

3.5.2 Class Distribution Analysis

Class distribution analysis confirmed a substantial imbalance: applying the BSMAS ≥ 19 cut-off classified approximately 11% of participants as "at-risk," consistent with the deliberately conservative nature of clinical cut-offs. Because a trivial classifier predicting "not at-risk" for every participant would already achieve approximately 89% accuracy, accuracy alone was deemed an insufficient/misleading performance metric for this study (see Section 3.7.3), and three imbalance-handling strategies were compared explicitly rather than applied provisionally.

3.5.3 Correlation Analysis

A correlation matrix was computed to examine the relationship between predictors and target. Heatmaps will be used to visualize multicollinearity. Correlations between individual predictors and the binary target were uniformly weak (|r| below 0.3).

3.6 Model Development

Supervised machine learning models were developed to classify individuals into addiction risk categories (e.g., low, moderate, high). The models estimate a function

$$f: R^p \rightarrow \{0,1, \ldots, K\}$$

where p represents the number of predictor variables and K represents the addiction risk classes.

3.6.1 Logistic Regression

Logistic Regression will serve as the baseline classifier due to its interpretability, statistical robustness, and probabilistic output. It models the probability of class membership using the logistic (sigmoid) function.

For binary classification:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}}$$

Where:

- β_0 = intercept
- β_i= coefficients
- x_i = predictor variables

The model estimates parameters by maximizing the likelihood function:

$$L(\beta) = \prod_{i=1}^{n} P(y_i \mid x_i)$$

3.6.2 Decision Trees

Decision Trees are non-parametric supervised learning models that recursively partition the feature space into homogeneous regions. They classify observations by learning decision rules inferred from the data features. At each node, the algorithm selects the feature that maximizes impurity reduction. For classification, impurity can be measured using:

Gini Index:

$$G = 1 - \sum_{k=1}^{K} p_k^2$$

or Entropy:

$$H = -\sum_{k=1}^{K} p_k \log_2 p_k$$

The optimal split is chosen by maximizing Information Gain:

$$IG = H(\text{parent}) - \sum_{i=1}^{m} \frac{N_i}{N} H(\text{child}_i)$$

Where:

- p_k = proportion of samples in class k
- N_i = number of samples in child node i
- N = total samples in parent node

Decision Trees are interpretable and capable of modeling nonlinear relationships, though they may overfit if not properly pruned.

3.6.3 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees using bootstrap sampling and random feature selection. It captures nonlinear relationships and reduces overfitting through aggregation. Each tree is trained on a bootstrap sample of the data. At each split, a random subset of features mmm is selected from the total p features.

The final prediction for classification is determined by majority voting:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \ldots, T_B(x))$$

Where:

- $T_b(x)$= prediction from the bth tree
- B = total number of trees

Feature importance can be computed using Mean Decrease in Impurity (MDI):

$$\text{Importance}(j) = \sum_{t \in \text{splits on } j} \Delta I_t$$

Where ΔI_t represents the reduction in impurity (e.g., Gini index) from splits involving feature j.

The Gini impurity measure is:

$$G = 1 - \sum_{k=1}^{K} p_k^2$$

3.6.4 Support Vector Machine (SVM)

Support Vector Machine aims to find the optimal hyperplane that maximizes the margin between classes.

For linearly separable data, the decision function is:

$$f(x) = w^T x + b$$

The objective is to minimize:

$$\min_{w,b} \frac{1}{2} || w ||^2$$

Subject to:

$$y_i(w^T x_i + b) \geq 1$$

For non-linearly separable data, slack variables ξi\xi_iξi are introduced:

$$\min_{w,b,\xi} \frac{1}{2} || w ||^2 + C \sum_{i=1}^{n} \xi_i$$

Subject to:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$$

Kernel functions transform data into higher-dimensional space:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Common kernels include linear, polynomial, and radial basis function (RBF):

$$K(x_i, x_j) = \exp(-\gamma \mid\mid x_i - x_j \mid\mid^2)$$

SVM is effective in high-dimensional spaces and performs well when class separation margins are distinct.

3.6.5 Handling of Class Imbalance

Given the 89/11 class split, three imbalance-handling strategies were compared for every classifier rather than assuming: (i) no rebalancing, (ii) class_weight='balanced' , which reweights the loss function inversely to class frequency, and (iii) SMOTE (Synthetic Minority Oversampling Technique, with k_neighbors=3 given the small minority count), which generates synthetic minority-class examples. Each combination of model × strategy (4 models × 3 strategies = 12 combinations) was fit as a single pipeline so that rebalancing was applied only within training folds, never leaking into validation or test data. A comprehensive overview of the hyperparameter search space used in tuning the models is presented in Table 2.

Table 2 Hyperparameter search space used for tuning the machine learning classification models during grid search cross-validation.

Model	Parameter	Values Tested	Purpose
Logistic Regression	C	0.01, 0.1, 1, 10, 100	Regularization strength
	Penalty	L2	Regularization type
	Solver	lbfgs, liblinear	Optimization algorithm
	Class Weight	None, balanced	Class imbalance handling
Decision Tree	Criterion	gini, entropy	Split quality metric
	Max Depth	3, 5, 7, 10, None	Tree depth
	Min Samples Split	2, 5, 10, 15	Minimum samples to split
	Min Samples Leaf	1, 2, 4, 6	Minimum samples per leaf
	Class Weight	None, balanced	Class imbalance handling
Random Forest	n_estimators	100, 200	Number of trees
	Max Depth	5, 8, 10, 15, None	Tree depth
	Min Samples Split	2, 5, 10	Minimum samples to split

	Min Samples Leaf	1, 2, 4	Minimum samples per leaf
	Max Features	sqrt, log2	Features per split
	Class Weight	None, balanced	Class imbalance handling
SVM	C	0.1, 1, 10, 50, 100	Regularization strength
	Kernel	linear, rbf	Kernel function
	Gamma	scale, auto, 0.01, 0.1, 1	Kernel coefficient
	Class Weight	None, balanced	Class imbalance handling

3.7 Model Evaluation

To ensure the robustness, stability, and generalizability of the proposed classification models, a comprehensive evaluation framework will be adopted. The assessment will focus on both out-of-sample predictive performance and class-level discrimination ability using multiple validation strategies and performance metrics.

3.7.1 Train/Test Split

The dataset was partitioned into training (80%) and testing (20%) subsets using a stratified split to preserve the 89/11 class ratio in both partitions. The test set was set aside and not used again for any model selection or tuning decision; it was reserved exclusively for one final, confirmatory evaluation of the selected model.

Let the full dataset be represented as:

$$D = \{(x_i, y_i)\}_{i=1}^{n}$$

where $x_i \in R^p$ represents the feature vector and $y_i \in \{0,1\}$ (or class labels) represents the target variable.

The model $f(\cdot)$ is trained on the training subset D_{train} and evaluated on the unseen testing subset D_{test}.

The generalization error is computed as:

$$\text{Test Error} = \frac{1}{|\, D_{test} \,|} \sum_{(x_i, y_i) \in D_{test}} L(y_i, f(x_i))$$

where $L(\cdot)$ denotes a suitable loss function (e.g., 0–1 loss for classification).

This approach provides an estimate of the model's ability to perform on new, unseen data.

3.7.2 Nested, Repeated, Stratified Cross-Validation

To avoid the optimistic bias that results from tuning hyperparameters and estimating performance on the same folds, a nested cross-validation design was adopted. An inner 5-fold stratified grid search tuned each model's hyperparameters (scored on PR-AUC, appropriate under class imbalance), while an outer repeated stratified 5-fold cross-validation (2 repeats, 10 total folds) estimated the generalization performance of each tuned pipeline. This procedure was applied to all 12 model × imbalance-strategy combinations.

3.7.3 Evaluation Metrics

Model performance will be evaluated using multiple classification metrics derived from the confusion matrix.

i. Accuracy

Accuracy measures the proportion of correctly classified instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- TP = True Positives
- TN = True Negatives
- TN= False Positives
- FN = False Negatives

ii. Precision

Precision measures the proportion of predicted positive cases that are truly positive:

$$\text{Precision} = \frac{TP}{TP + FP}$$

It reflects the model's ability to avoid false alarms.

iii. Recall (Sensitivity)

Recall measures the proportion of actual positive cases correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

It indicates the model's ability to detect positive instances.

iv. F1-Score

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric is particularly useful when dealing with imbalanced datasets.

v. ROC-AUC

The ROC curve plots:

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP + FN}$$

against

$$\text{False Positive Rate (FPR)} = \frac{FP}{FP + TN}$$

at various classification thresholds.

The Area Under the Curve (AUC) is defined as:

$$AUC = \int_0^1 TPR(FPR)\, d(FPR)$$

AUC values range from 0 to 1, where:

- 0.5 indicates random performance,
- 1.0 indicates perfect classification.

ROC curves will be plotted to compare the discriminative ability of all models.

vi. PR-AUC (Average Precision)

Because accuracy is a poor discriminator of performance under severe class imbalance; a classifier predicting the majority class for every case already scores approximately 89% accuracy here the area under the precision–recall curve was used as the primary model-selection criterion, alongside recall and F1 on the positive ("at-risk") class.

vii. Baseline Comparison

A DummyClassifier (predicting the most frequent class, and a stratified-random variant) was evaluated on the same metrics to establish the minimum performance any informative model must exceed.

3.7.4 Confusion Matrix

The confusion matrix provides a detailed breakdown of classification outcomes:

$$\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}$$

It enables the identification of:

- Correct classifications (TP, TN),
- Type I errors (FP),
- Type II errors (FN).

For multi-class classification, the confusion matrix generalizes to a $C \times C$ matrix, where C represents the number of classes.

3.8 Explainable AI

To ensure transparency and interpretability in the behavioral analysis, the study will utilize Explainable Artificial Intelligence (XAI) techniques, moving beyond "black-box" predictions to understand the underlying drivers of addiction risk. SHAP (SHapley Additive exPlanations) values, grounded in game theory, will be the primary tool used to quantify the contribution of each feature to the model's output. This includes examining global feature importance to identify which predictors are most influential across the entire population, alongside local

explanations that provide granular insights into why the model generated a specific risk score for an individual participant.

3.9 Assumptions

The methodology adopted in this research is guided by the following assumptions:

i. It is assumed that the dataset used for this study contains accurate, reliable, and truthfully reported responses from participants regarding their social media usage patterns and behavioral indicators.

ii. The dataset is assumed to be reasonably representative of the broader population under study, such that patterns learned by the models can generalize beyond the sample.

iii. Each record in the dataset is assumed to represent an independent individual, with no duplicate or dependent entries that could bias model learning.

3.10 Limitations

Despite methodological rigor, this study is subject to the following limitations:

i. Since the dataset is based on self-reported responses, it may contain recall bias or socially desirable responding, potentially affecting prediction accuracy.

ii.The dataset is geographically or demographically constrained, findings may not fully generalize to different populations or cultural contexts.

ii. Training complex models with large hyperparameter spaces may require significant computational time and resources, potentially limiting extensive hyperparameter optimization.

iv. The severe class imbalance (approximately 89% not-at-risk vs. 11% at-risk, with only about 23 at risk cases in the training set) limits how much the minority class can be learned, even with class-weighting or SMOTE; SMOTE's synthetic oversampling in particular assumes the minority class's feature space is well-represented enough to support interpolation, which is a tenuous assumption at this sample size.

v. The small overall sample size (n = 258) constrains the extent of hyperparameter tuning and the precision of the nested cross-validation performance estimates (reflected in the standard deviations reported alongside mean scores).

4. RESULTS AND DISCUSSION

4.1. Introduction

This chapter presents the empirical findings obtained by applying the methodology described in Chapter Three to the Social Media Addiction dataset (N = 258). The chapter is organised to mirror the analytical pipeline set out in the previous chapter: it begins with exploratory data analysis (EDA) covering sample characteristics, the distribution of Bergen Social Media Addiction Scale (BSMAS) scores, platform-usage patterns, and correlational structure; it then reports the construction of the binary "at-risk" target and the class-imbalance diagnostics that follow from it; it presents the comparative performance of four classifiers (Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine) combined with three imbalance-handling strategies, evaluated through nested, repeated, stratified cross-validation; it reports the final held-out test evaluation of the selected model; and it closes with the SHAP-based explainability analysis. Each subsection is discussed against the two competing evaluation lenses established methodologically ranking ability (PR-AUC) versus decision-threshold performance (F1) given the pronounced class imbalance in the dataset.

4.2. Sample Characteristics

The raw file contained 258 rows and 43 columns; after removing duplicate columns, 25 columns remained, of which 24 substantive variables (23 predictors plus the BSMAS total score) were retained for analysis. No missing values were identified in any of the 24 retained variables, so no listwise deletion or imputation was required at this stage. Table 3 shows the summary of the sample demographic.

Table 3 Summary of sample demographic characteristics (N = 258)

Characteristic	Statistic	Value
Sample size	N	258
Age (years)	Mean (SD)	17.42 (1.73)
Age (years)	Range (Min–Max)	14 – 22
Sex	Male, n (%)	109 (42.2%)
Sex	Female, n (%)	149 (57.8%)

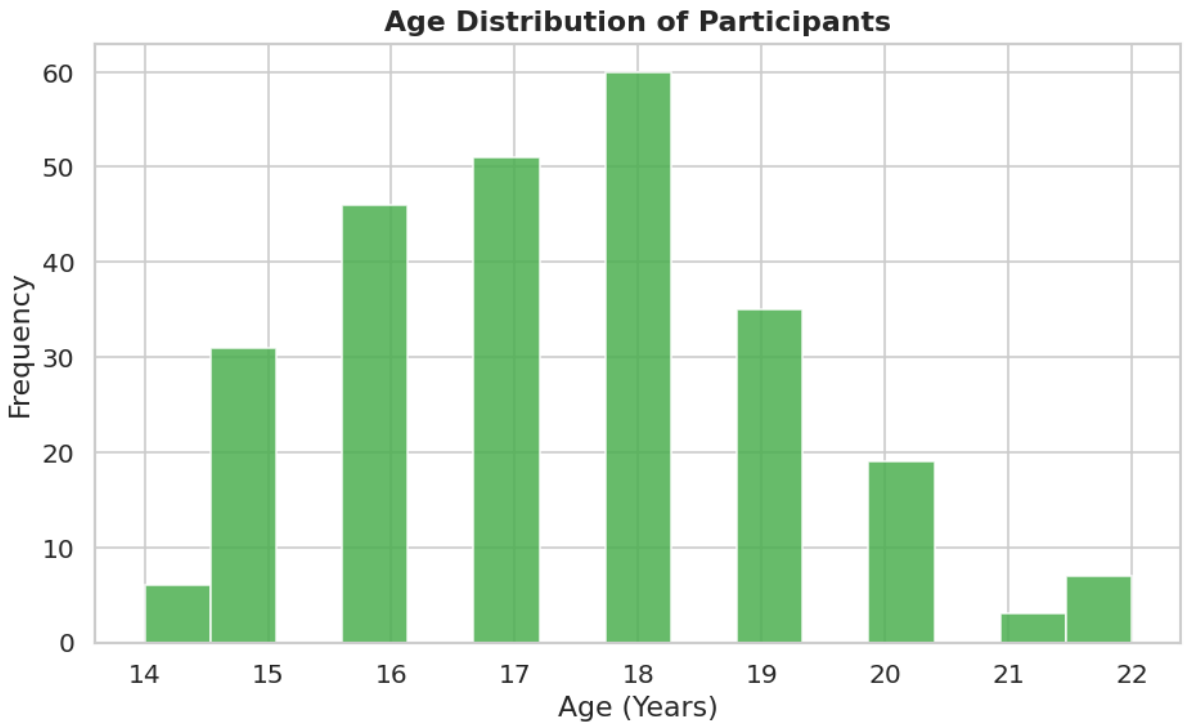


Figure 3 Age distribution of participants

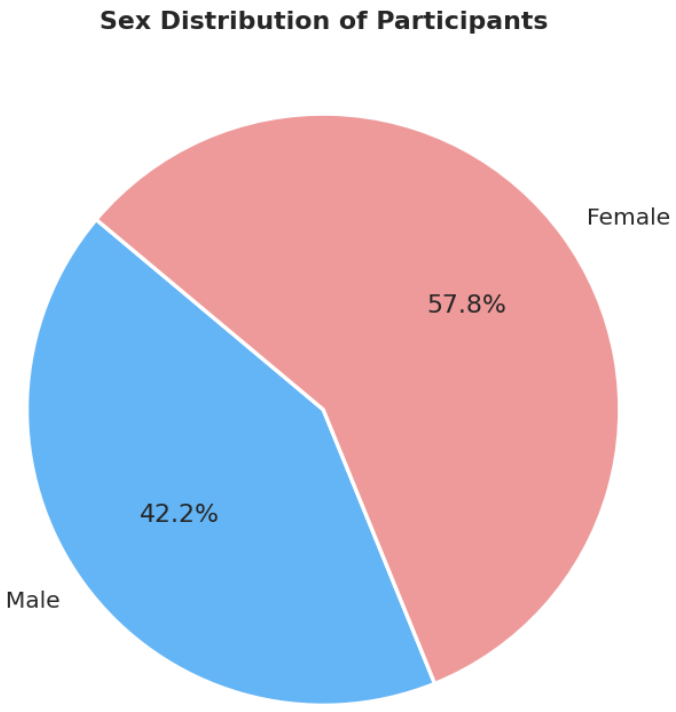


Figure 4 Sex distribution of participants

The sample is composed mainly of adolescents and young adults (mean age = 17.4 years, SD = 1.73), with a slightly higher proportion of female participants (57.8%) than male participants (42.2%). This distribution is broadly consistent with the target population described in the methodology and provides an adequate demographic spread for examining age and sex related patterns in addictive social media use, although the modest sample size (N = 258) should be kept in mind when interpreting subgroup comparisons

4.3 Distribution of Addiction Scores

The BSMAS score, from which the binary addiction-risk target is derived, ranges from 6 to 27 (Mean = 13.07, Median = 13.00, SD = 4.48). The closeness of the mean and median indicates a distribution that is roughly symmetric rather than heavily skewed, which is visually confirmed in Figure 5.

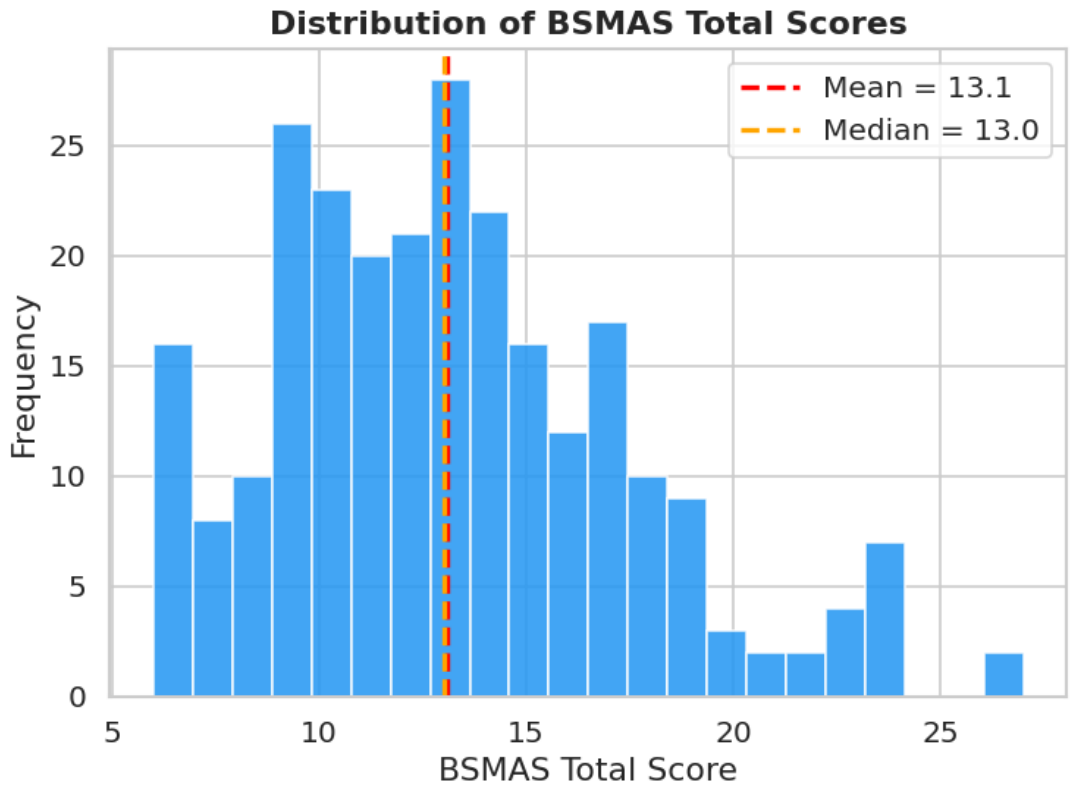


Figure 5 Distribution of BSMAS total scores

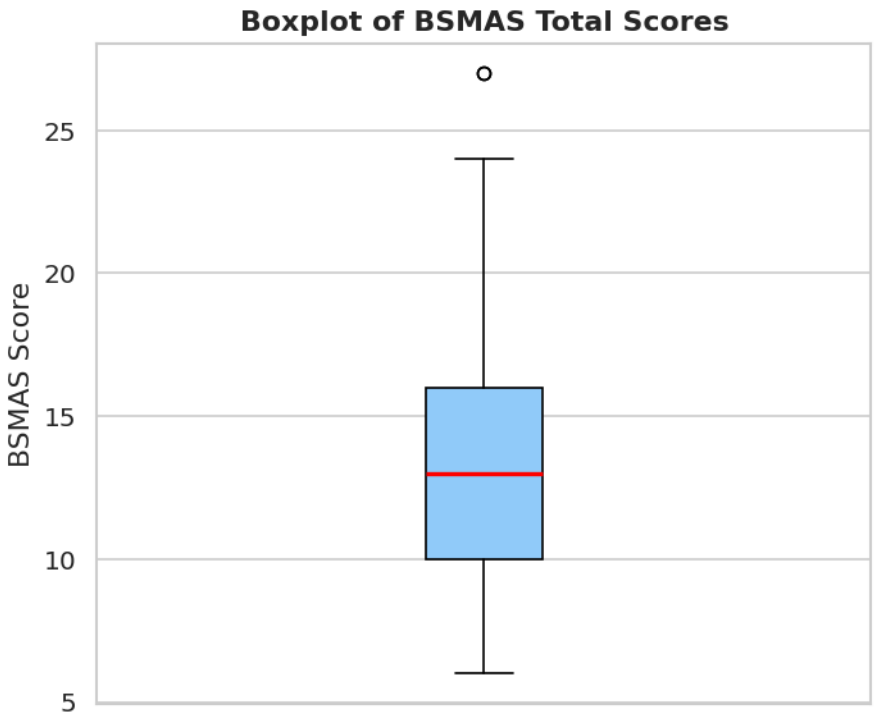


Figure 6 Boxplot of BSMAS total scores

The boxplot in Figure 6 shows a moderate spread of scores around the median with a single higher-scoring outlier, consistent with the small proportion of participants who exceed the clinical at-risk cut-off.

4.4 Social Media Platform Usage

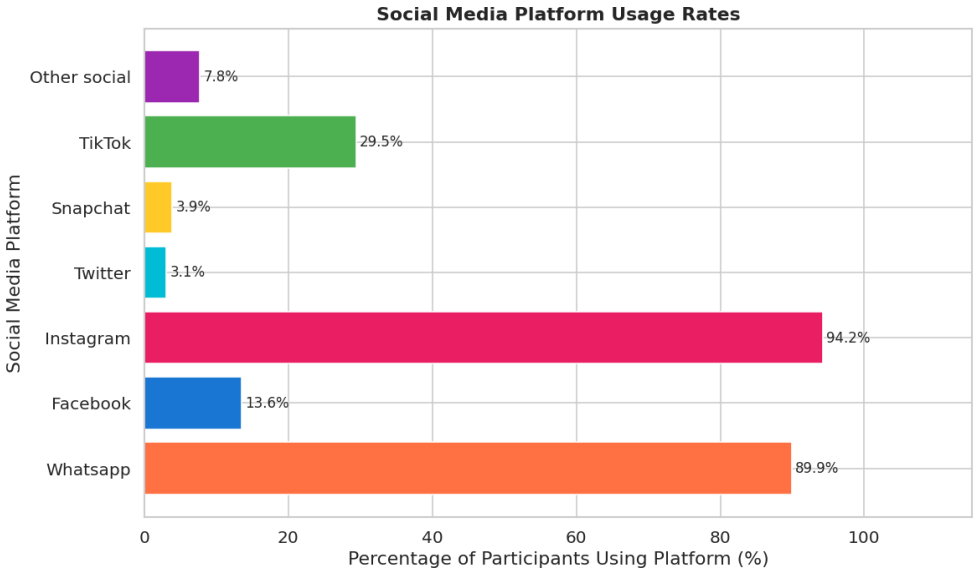


Figure 7 Social media platform usage rates across participants

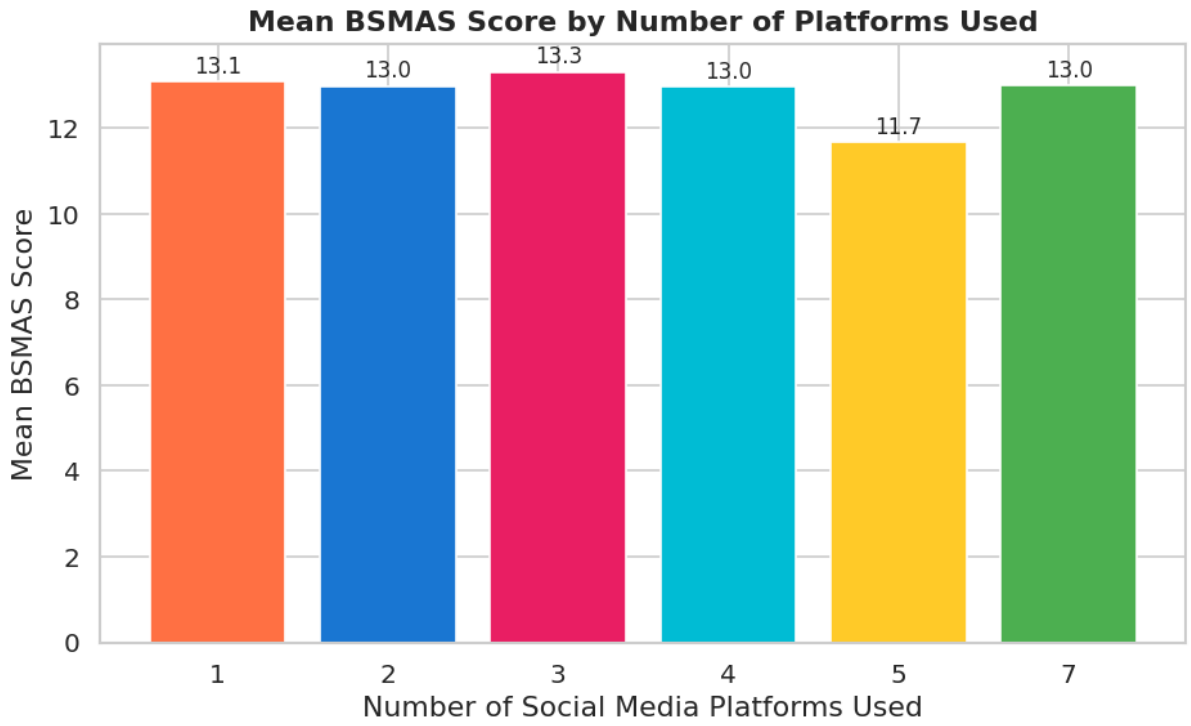


Figure 8 Mean BSMAS score by number of social media platforms used

Figure 7 summarises the proportion of participants using each of the seven platform categories captured in the dataset (WhatsApp, Facebook, Instagram, Twitter, Snapchat, TikTok, and Other). Figure 8 aggregates participants by the total number of distinct platforms they report using (the engineered n_platforms feature) and plots the corresponding mean BSMAS score for each group. This engineered feature was carried forward into the modelling stage as a predictor.

4.5 The Target

Applying the validated BSMAS clinical cut-off (score ≥ 19) produced a binary target with a great class imbalance: 229 participants (88.8%) were classified as "Not at-risk" and only 29 participants (11.2%) were classified as "At-risk" (Table 4, Figure 9). This split has direct consequences for model evaluation, since a classifier that trivially predicts "Not at-risk" for every participant would already achieve close to 89% accuracy without learning anything about addiction risk. This motivated the use of PR-AUC, positive-class F1/recall, and an explicit dummy-classifier baseline throughout the remainder of the chapter, rather than relying on accuracy alone.

Table 4 Distribution of the binary addiction-risk target (BSMAS cut-off ≥ 19)

Class	n	% of sample
Not at-risk (BSMAS < 19)	229	88.8%
At-risk (BSMAS ≥ 19)	29	11.2%
Total	258	100.0%

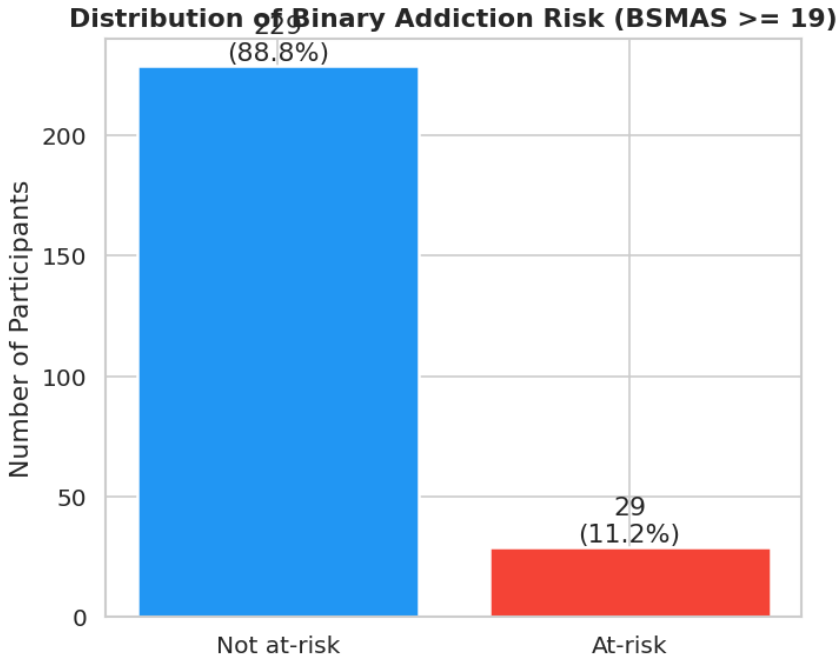


Figure 9 Distribution of binary addiction-risk classes

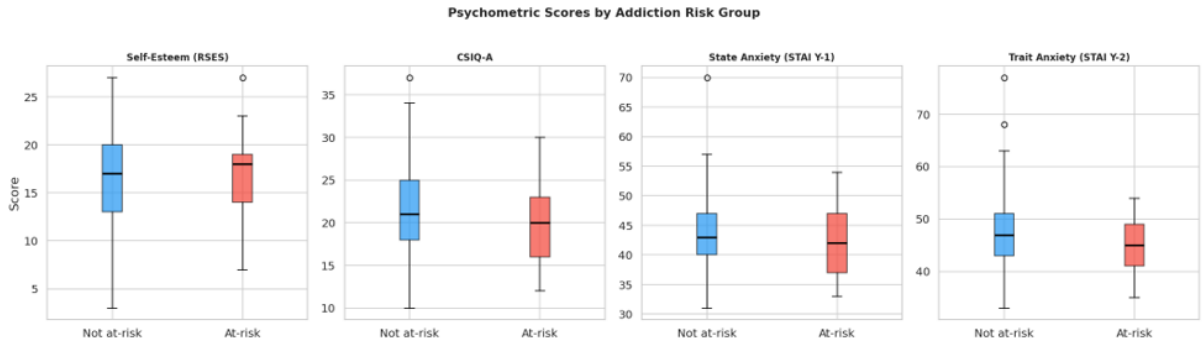


Figure 10 Psychometric scores (RSES, CSIQ-A, STAI Y-1, STAI Y-2) compared across at-risk and not-at-risk groups

Figure 10 compares self-esteem (RSES), CSIQ-A, state anxiety (STAI Y-1), and trait anxiety (STAI Y-2) scores between the two risk groups. These four psychometric variables reappear as the most influential predictors in the SHAP explainability analysis, which lends some support to the descriptive pattern shown here even though the group-level separation is not clear in terms of raw correlation.

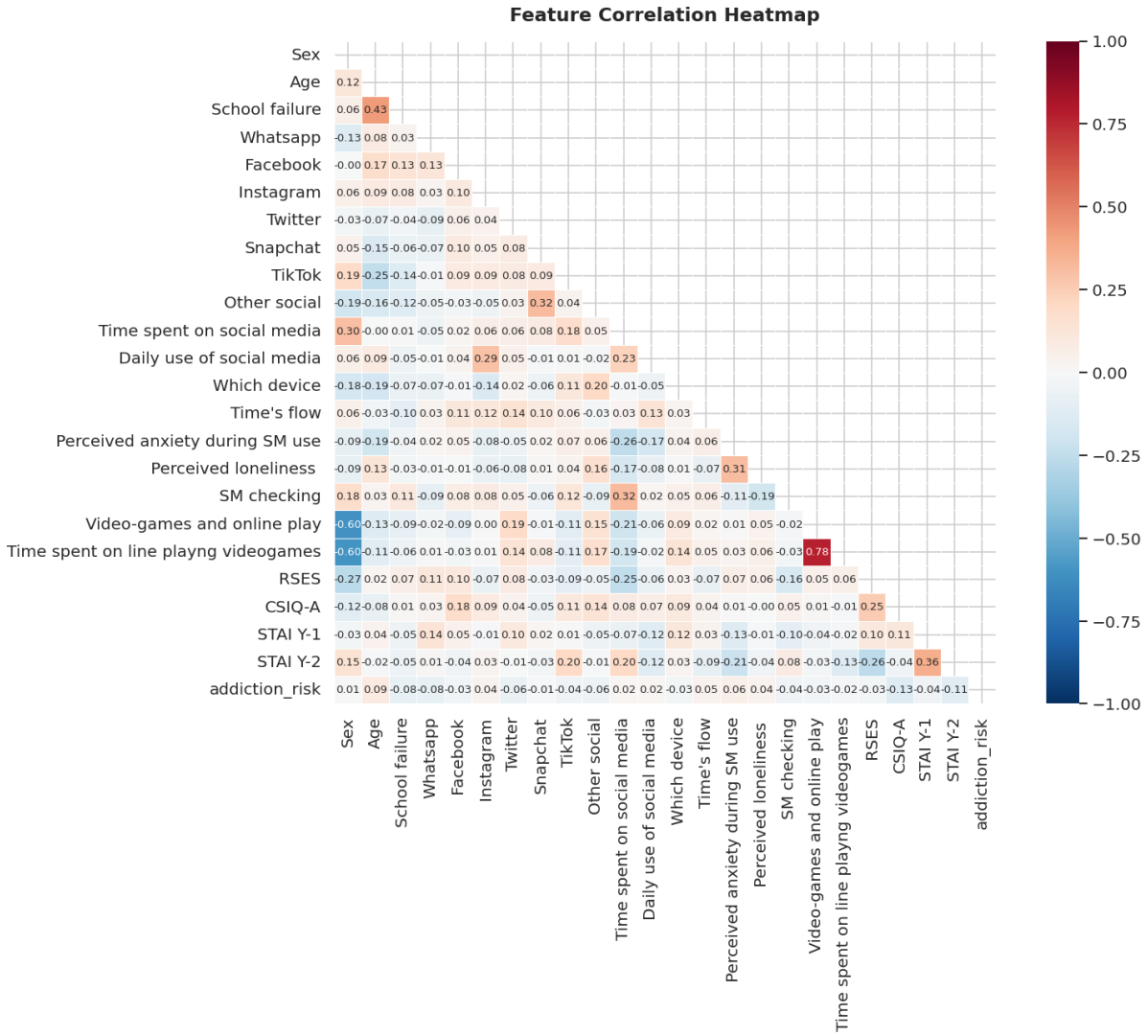


Figure 11 Correlation heatmap of all numeric predictors and the addiction-risk target

Table 5 lists ten features with the strongest absolute linear correlation to the target. In absolute terms all correlations are weak (|r| below 0.2 for every feature) with CSIQ-A (r = -0.132) and trait anxiety, STAI Y-2 (r = -0.113) showing the strongest associations.

Table 5 Ten features with the strongest correlation with the binary addiction-risk target

Feature	Correlation with addiction_risk (r)
CSIQ-A	-0.132
STAI Y-2 (Trait Anxiety)	-0.113
Age	0.091
Whatsapp (platform use)	-0.085
School failure	-0.083
Twitter (platform use)	-0.064
Other social (platform use)	-0.057
Perceived anxiety during SM use	0.057

Time's flow	0.046
Perceived loneliness	0.045

4.6 Feature Preparation and Train/Test Split

One-hot encoding of the two nominal categorical variables (Sex and Which device) was performed as well as z-score standardisation of all remaining numeric predictors. The final feature matrix comprised 22 numeric features and 2 categorical variables. The dataset was then partitioned into an 80% training set (n = 206) and a 20% held-out test set (n = 52) using stratified sampling, which preserved approximately 89:11 class balance in both partitions.

4.7 Baseline Performance

Before evaluating any of the four candidate classifiers, two naive DummyClassifier baselines were run on the held-out test set to establish what "doing nothing clever" scores under this class balance (Table 6). As anticipated, the “most frequent” strategy (always predicting "Not at-risk") achieved 88.5% accuracy while identifying zero at-risk cases (F1 = 0.0 on the positive class), confirming that accuracy alone cannot distinguish a genuinely predictive model from one that has learned nothing. Both baselines scored a PR-AUC of 0.115, which is simply the base rate of the positive class and forms the floor against which every trained model's PR-AUC should be judged.

Table 6 Baseline Performance

Strategy	Accuracy	F1 (at-risk)	PR-AUC	ROC-AUC
most_frequent	0.885	0.000	0.115	0.500
stratified	0.769	0.000	0.115	0.435

4.8 Comparative Model Performance

Four classifiers were each paired with three imbalance-handling strategies (no rebalancing, class_weight = 'balanced', and SMOTE), producing 12 model/strategy combinations. Each combination was evaluated using nested, repeated, stratified cross-validation: an inner 5-fold grid search tuned hyperparameters within each outer training fold, and an outer 5-fold × 2-repeat (10-fold) stratified split estimated out-of-sample performance of the tuned pipeline. Table 7 reports the mean (± SD) performance of all 12 combinations, ranked by PR-AUC.

Table 7 Nested cross-validation results for all 12 model × imbalance-strategy combinations

Model	Balancing	Accuracy	F1	PR-AUC	ROC-AUC
Random Forest	none	0.862 ± 0.049	0.104 ± 0.137	0.232 ± 0.095	0.517 ± 0.180
Random Forest	class_weight	0.862 ± 0.049	0.104 ± 0.137	0.232 ± 0.095	0.517 ± 0.180
SVM	smote	0.813 ± 0.101	0.090 ± 0.123	0.231 ± 0.093	0.547 ± 0.122
Random Forest	smote	0.840 ± 0.034	0.108 ± 0.109	0.214 ± 0.092	0.506 ± 0.150
Logistic Regression	smote	0.643 ± 0.079	0.137 ± 0.103	0.192 ± 0.063	0.521 ± 0.116
Decision Tree	none	0.751 ± 0.102	0.215 ± 0.178	0.189 ± 0.086	0.578 ± 0.146
Decision Tree	class_weight	0.751 ± 0.102	0.215 ± 0.178	0.189 ± 0.086	0.578 ± 0.146
Logistic Regression	none	0.707 ± 0.118	0.158 ± 0.103	0.176 ± 0.053	0.498 ± 0.113
Logistic Regression	class_weight	0.707 ± 0.118	0.158 ± 0.103	0.176 ± 0.053	0.498 ± 0.113
SVM	none	0.802 ± 0.118	0.035 ± 0.054	0.162 ± 0.060	0.459 ± 0.105
SVM	class_weight	0.802 ± 0.118	0.035 ± 0.054	0.162 ± 0.060	0.459 ± 0.105
Decision Tree	smote	0.724 ± 0.081	0.100 ± 0.095	0.123 ± 0.030	0.475 ± 0.094

Ranking the 12 model–balancing combinations by PR-AUC places Random Forest (with either no rebalancing or class_weight = 'balanced', which produced identical results) at the top, achieving a PR-AUC of 0.232 ± 0.095, followed closely by SVM with SMOTE (0.231 ± 0.093). Both models perform substantially better than the baseline PR-AUC of 0.115 established in Section 4.7, indicating an improved ability to distinguish at-risk users across decision thresholds. However, ranking the same combinations by F1 score produces a different ordering. Decision Tree (with no rebalancing or class_weight = 'balanced') achieves the highest F1 score (0.215 ± 0.178), despite a lower PR-AUC (0.189 ± 0.086) than Random Forest. This divergence reflects the trade-off between PR-AUC and F1 identified in the methodology. Random Forest provides better overall ranking of high-risk users across all classification thresholds, resulting in the highest PR-AUC, but predicts relatively few positive cases at the default decision threshold, yielding a lower F1 score (0.104 ± 0.137). In contrast, the Decision Tree identifies more high-risk users at the default threshold, leading to a higher F1 score despite weaker ranking performance overall.

4.9 Final Model Selection and Held-out Test Evaluation

Ranking by PR-AUC, Random Forest with no explicit rebalancing ("none") was selected as the final model. The internally optimal hyperparameters reported in Table 8 includes class_weight = 'balanced', even though the pipeline-level imbalance strategy was labelled "none" (i.e., no SMOTE oversampling step was used). This is not a contradiction; the imbalance-strategy comparison in Section 4.8 concerns whether a resampling step (SMOTE) is added to the pipeline, while class_weight remains an independent hyperparameter for the Random Forest model itself.

Table 8 Best hyperparameters for the selected model (Random Forest), refit on the full training set

Hyperparameter	Value
class_weight	balanced
max_depth	5
max_features	sqrt
min_samples_leaf	1
min_samples_split	2
n_estimators	100

The selected model was then evaluated once on the untouched held-out test set (n = 52). Overall accuracy was 0.885, but as mentioned earlier in the chapter, this value is the same as that achieved by most_frequent baseline (See Table 6). The more decision-relevant metrics obtained: positive-class (at-risk) precision was 0.500, recall was only 0.167, and F1 was 0.250. PR-AUC (0.361) and ROC-AUC (0.676) both exceed the respective baseline values (0.115 and 0.500). This shows that the model ranks at-risk cases somewhat better than chance, even though it identifies only 1 of the 6 at-risk participants in the test set at the default 0.5 threshold (Table 9, Figure 12).

Table 9 Final Held-out Test Performance

Class	Precision	Recall	F1-score	Support
Not at-risk	0.90	0.98	0.94	46
At-risk	0.50	0.17	0.25	6
Accuracy			0.88	52
Macro avg	0.70	0.57	0.59	52
Weighted avg	0.85	0.88	0.86	52

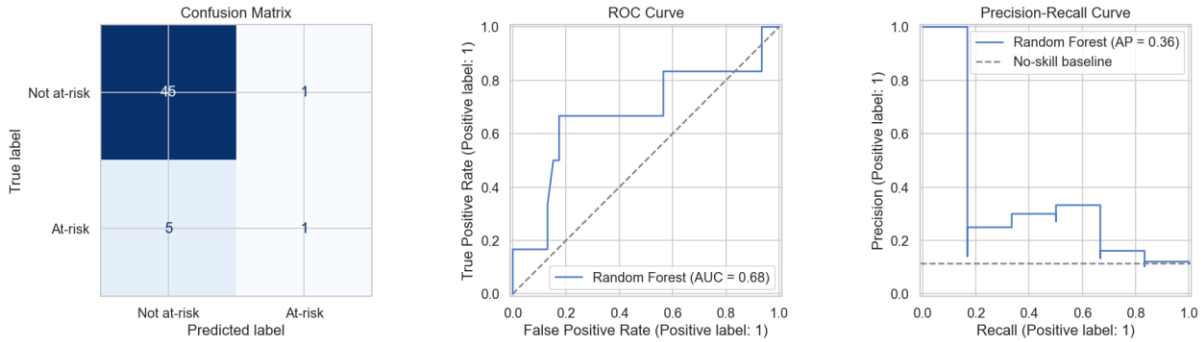


Figure 12 Confusion matrix, ROC curve, and precision-recall curve for the final model on the held-out test set

4.10 Model Explainability

SHAP (SHapley Additive exPlanations) values were computed for the final Random Forest model using a tree-based explainer, applied to the held-out test set. Table 10 lists the ten features with the highest mean absolute SHAP value, i.e. the features the model relies on most when producing its at-risk predictions.

Table 10 Top 10 Features by Mean

Rank	Feature	Mean |SHAP value|
1	CSIQ-A	0.0619
2	STAI Y-2 (Trait Anxiety)	0.0434
3	RSES (Self-Esteem)	0.0391
4	STAI Y-1 (State Anxiety)	0.0284
5	School failure	0.0280
6	n_platforms (engineered)	0.0183
7	Perceived anxiety during SM use	0.0172
8	Age	0.0151
9	Time spent on social media	0.0127
10	Video-games and online play	0.0094

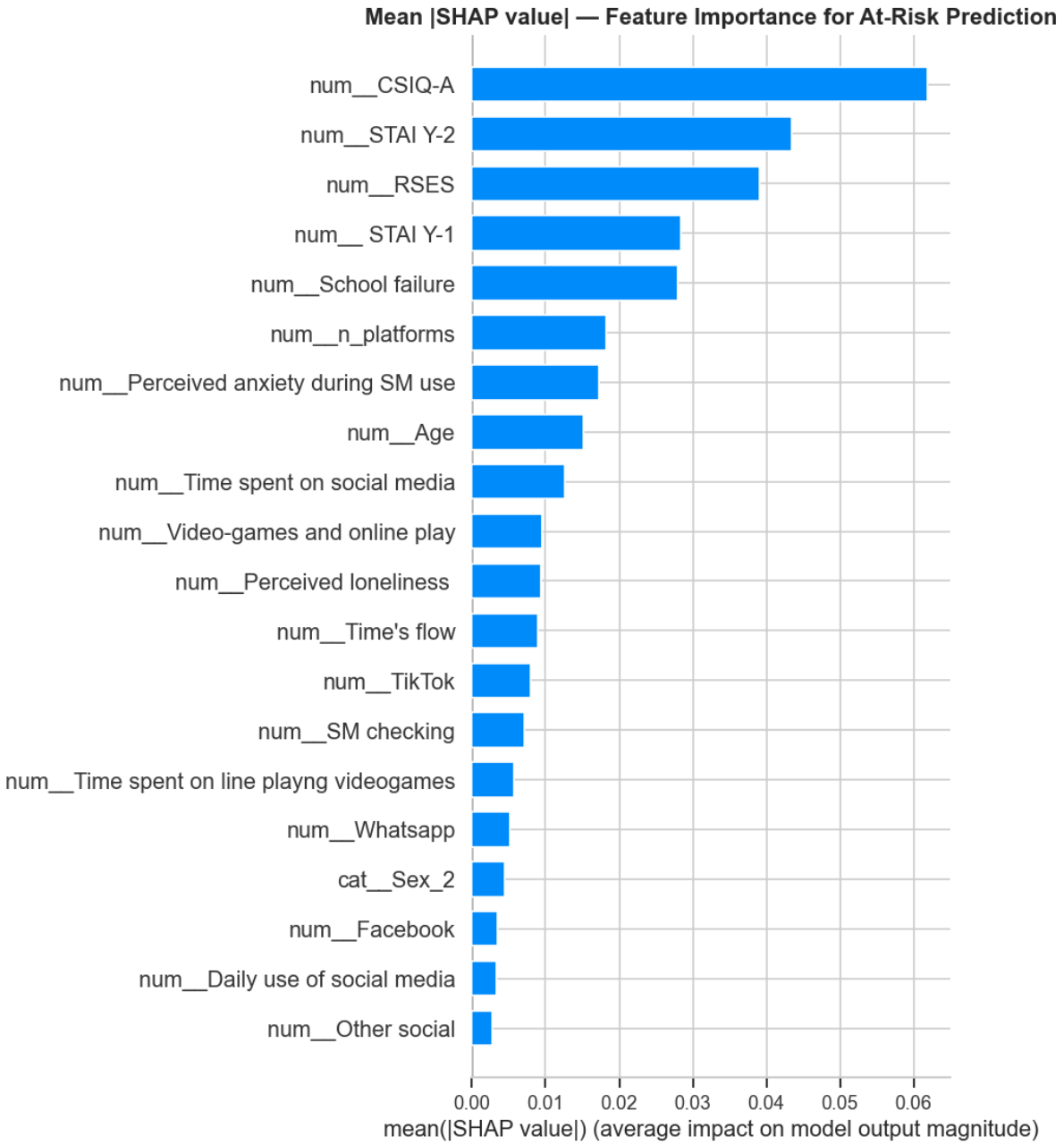


Figure 13 Mean feature importance ranking for at-risk prediction

The SHAP ranking is dominated by the psychometric variables: CSIQ-A, trait anxiety (STAI Y-2), self-esteem (RSES), and state anxiety (STAI Y-1) occupy the top four positions. This aligns with the heatmap shown in Figure 11 and the correlation ranking in Table 5, even though none of these individual relationships is strong in linear-correlation terms. It suggests these psychometric features carry more information about addiction-risk status than the behavioural/platform-usage variables, even though the overall predictive power of the model remains modest.



Figure 14 SHAP summary (beeswarm) plot showing the direction and magnitude of each feature's contribution

5. SUMMARY AND CONCLUSION

5.1 Introduction

This chapter brings together the findings reported in the previous chapter and considers them against the aim, research questions, and objectives set out in Chapter One. It begins with a summary of what the study set out to do and how it was carried out, before turning to the conclusions that can reasonably be drawn from the results. The chapter then reflects on the contribution of the study, revisits its limitations considering the results obtained, and closes with recommendations for practice and for future research.

5.2 Summary of the Study

This study set out to develop and evaluate predictive and explainable machine learning models for identifying social media addiction risk, using a publicly available dataset of 258 adolescents and young adults. Two research questions guided the work: whether machine learning models could accurately predict social media addiction risk, and which variables contribute most significantly to those predictions. Four objectives followed from these questions: to identify key behavioural and demographic factors associated with addiction; to develop predictive models using both traditional statistical and machine learning techniques; to compare model performance using appropriate classification metrics; and to apply explainable AI techniques to interpret the resulting predictions.

To address these objectives, the Bergen Social Media Addiction Scale (BSMAS) total score was used to construct a binary "at-risk" target using a validated clinical cut-off, which produced a markedly imbalanced dataset (88.8% not-at-risk, 11.2% at-risk). Four classifiers, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine, were each combined with three imbalance-handling strategies (no rebalancing, class-weighting, and SMOTE), giving twelve candidate pipelines. These were evaluated using nested, repeated, stratified cross-validation, with model ranking assessed against both PR-AUC, which reflects overall ranking ability, and F1, which reflects performance at a fixed decision threshold. The best-performing configuration by PR-AUC, a Random Forest model, was retrained on the full training set and evaluated once on a held-out test set. SHAP was then applied to this final model to identify which features it relied on most heavily when generating predictions

5.3 Summary of Key Findings

Three findings stand out from the results reported in Chapter Four. First, the study established a clear evaluation baseline before drawing any conclusions about model performance given the 89:11 class split, a classifier that predicted "not at-risk" for every participant would achieve close to 89% accuracy while identifying no at-risk cases at all. Both dummy baselines produced a PR-AUC of 0.115, the base rate of the positive class, and it is this figure, rather than accuracy, that the trained models needed to exceed for any claim of genuine predictive value to hold.

Second, model comparison revealed a genuine but modest improvement over this baseline, and a trade-off between two reasonable definitions of "best." Ranked by PR-AUC, Random Forest performed best (0.232 ± 0.095), narrowly ahead of SVM with SMOTE (0.231 ± 0.093), both roughly double the baseline PR-AUC. Ranked by F1 score at the default threshold, however, the Decision Tree performed best (0.215 ± 0.178) despite a lower PR-AUC, because it identified more positive cases at that threshold even though it ranked cases less reliably overall. When the selected Random Forest model was evaluated on the held-out test set, it achieved a PR-AUC of 0.361 and ROC-AUC of 0.676, both clearly above chance, but at the default 0.5 threshold it identified only one of the six at-risk participants in the test set (recall = 0.167, precision = 0.500, F1 = 0.250). Taken together, these results indicate that the model can rank at-risk individuals meaningfully better than chance, but is not, in its current form, a reliable tool for flagging individual cases at a fixed threshold.

Third, the SHAP explainability analysis showed that the model's predictions were dominated by psychometric variables rather than social media usage variables. CSIQ-A, trait anxiety (STAI Y-2), self-esteem (RSES), and state anxiety (STAI Y-1) occupied the four highest-ranked positions by mean absolute SHAP value, ahead of behavioural and platform-related features such as school failure, the number of platforms used, perceived anxiety during use, age, time spent on social media, and gaming activity. This pattern is broadly consistent with the descriptive comparisons in Chapter Four, where these same psychometric variables separated the at-risk and not-at-risk groups more than any usage variable did, even though none of the individual linear correlations with the target exceeded $|r| = 0.2$.

5.4 Conclusions

Returning to the first research question, whether machine learning models can accurately predict social media addiction risk, the answer supported by this study is a qualified yes. The

best model exceeded the baseline PR-AUC by a factor of roughly three and achieved a ROC-AUC well above chance on unseen data, so the models did learn a signal that a naive classifier could not access. At the same time, as mentioned in chapter 4, the models accuracy needs to be read carefully: the low recall at the default threshold (0.167) means the model, as configured, missed most of the at-risk participants in the test set. This is not a contradiction so much as a reflection of the difficulty of the underlying problem, given the small number of at-risk cases (29 in total, roughly 23 in any given training fold) from which the model had to learn a generalisable pattern. It is consistent with the wider methodological literature cautioning that prediction and explanation trade off against each other under exactly these small-sample, high-imbalance conditions (Yarkoni and Westfall, 2017; Lavelle-Hill, Smith and Murayama, 2025).

On the second research question, which variables contribute most to the predictions, the study concludes that psychometric indicators of internal distress, cognitive interference from smartphone use (CSIQ-A), anxiety, and low self-esteem, contribute more to the model's predictions than behavioural indicators of social media use itself, such as time spent, number of platforms, or specific platform choice. Much of the reviewed literature frames social media addiction primarily around usage patterns and platform design (e.g. intermittent reinforcement, notification-driven engagement), consistent with the components model of behavioural addiction (Griffiths, 2005). The present results suggest that, at least within this sample, pre-existing psychological vulnerability may be a stronger signal of addiction risk than the surface-level behaviour of how much or how widely a person uses social media. This does not contradict the behavioural-addiction framing, instead, it suggests where the more informative predictors sit within it.

More broadly, the study concludes that explainable machine learning offers a genuine, if modest, improvement over the correlation-based approaches that dominate the existing literature on social media addiction (Panova and Carbonell, 2022). The value of the SHAP analysis in this study was not that it uncovered dramatically different variables from those already implicated by prior descriptive and correlational work, but that it quantified their relative contribution to actual predictions in a way that traditional bivariate correlation could not. This study's conclusions are offered with appropriate caution given its sample size and imbalance constraints, and recommends for future studies, larger sample, longitudinal designs, and broader external validation as the most productive next steps for this line of research.

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